

QUERY-ADAPTIVE MONGE TANGENT TRANSFER FOR MULTI-SOURCE DISTRIBUTIONAL REGRESSION IN WASSERSTEIN SPACE

Anonymous authors

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ABSTRACT

Transfer learning for probability-distribution-valued responses residing in Wasserstein space is critical for leveraging auxiliary multi-cohort data in modern data science. However, existing state-of-the-art Wasserstein transfer learning frameworks suffer from two coupled pathologies: raw linear directional Fréchet weights frequently evaluate to negative values (occurring in $\sim 14.9\%$ of query evaluations), generating invalid signed measures that mandate unstable post-hoc monotone projections, while static global reference measures induce Monge-map geometric distortion under non-linear domain shift. We introduce *Query-Adaptive Monge Tangent Transfer* (QAMT-Transfer), a multi-source transfer learning framework for distribution-valued responses that unifies four algorithmic innovations: (1) convex exponential-tilting Fréchet–RKHS covariate weighting normalized to the probability simplex, guaranteeing valid probability measures without post-hoc optimization; (2) query-adaptive local target reference barycenters centered dynamically at each query point, preserving local manifold geometry; (3) tangent-space velocity field linearization within Hilbert spaces $L^2(\sigma_0(x))$; and (4) Sobolev H^1 -regularized continuous soft attention with closed-form push-forward shrinkage. Across all 8 univariate regimes, QAMT-Transfer achieves a consistent 11.7% average reduction in Root Mean Squared Prediction Risk (RMSPR) over state-of-the-art Wasserstein Transfer Learning, while preserving strictly valid and monotone quantile predictions by construction. On a multivariate ($d = 2$) benchmark with non-linear transport maps, the geometric stages (2–4) deliver a statistically significant additional gain over convex weighting alone ($p < 0.05$), and QAMT-Transfer reduces RMSPR by up to 20.1% over WaTL. A controlled dimension sweep ($d \in \{1, 2, 3, 5\}$) further shows that the geometric stages deliver a statistically significant 8–13% risk reduction under non-linear multivariate transport while cleanly collapsing to Stage-1 in 1-D, and QAMT-Transfer consistently outperforms recent adaptive (AWaTL) and neural (NLWR) transport baselines across 1-D and 2-D. On three real-world public benchmarks—Eurostat regional age-at-death distributions, NHANES physical-activity distributions, and a new 2-D NYC taxi spatial drop-off transfer task—QAMT-Transfer consistently outperforms WaTL, with up to a 53% error reduction under extreme target scarcity. Systematic ablations isolate the contribution of each stage across dimensions, and a GPU latency study confirms sub-second per-query inference for $d \geq 2$.

1 INTRODUCTION

Statistical analysis of complex, non-Euclidean random objects situated in metric spaces—such as probability distributions, correlation networks, and Riemannian manifolds—has emerged as a foundational domain in modern data science (Petersen & Muller, 2016; Le & Pham, 2023). In particular, analyzing conditional responses structured as probability measures within the 2-Wasserstein space $\mathcal{W}_2(\mathbb{R}^d)$ provides powerful analytical capabilities across demographic modeling, medical imaging, and financial risk assessment (Peyré & Cuturi, 2018; Ghodrati & Panaretos, 2021). However, in real-world scenarios where data collection in the target domain is severely constrained ($n_0 \leq 200$), classical single-domain regression frameworks suffer from high variance, parameter instability, and

poor out-of-sample generalization (Kurusu & Otsu, 2025). Multi-source transfer learning offers a promising paradigm to overcome target-data scarcity by pooling knowledge from multiple related auxiliary studies (Xiong et al., 2023; Feng et al., 2021).

Recently, Wasserstein Transfer Learning (WaTL) (Zhang et al., 2025) introduced multi-source transfer for distributional responses by computing directional linear Fréchet regression weights across auxiliary cohorts. Despite its theoretical promise, existing state-of-the-art distributional transfer methods suffer from two critical mathematical and geometric pathologies:

1. **Invalid measure extrapolations via unbounded linear weights.** Standard directional linear Fréchet weights $s_{iG}^{(k)}(x) = 1 + (X_i^{(k)} - \bar{X}_k)^\top \hat{\Sigma}_k^{-1}(x - \bar{X}_k)$ can become negative or unbounded for query points x situated far from sample means. Empirically, unconstrained linear weighting yields invalid, negative measure evaluations on approximately 14.9% of query evaluations. This invalidity destroys the Euclidean convexity of Wasserstein barycenter problems (Jacobs & Zhou, 2026; Zalles et al., 2024; Zhou & Uribe, 2026), requiring unstable, post-hoc monotone projections via operator-splitting quadratic programming (OSQP) solvers (Stellato et al., 2017).
2. **Geometric distortion from static global reference linearization.** Conventional optimal-transport linearizations project sample distributions into a tangent space constructed around a single, static global reference measure σ_0 (e.g., global target or pooled sample barycenters) (Wang et al., 2012; Bigot et al., 2013). When conditional distributions shift non-linearly across the covariate space, static linearizations induce high Monge-map non-smoothness and geometric distortion (Uscidda & Cuturi, 2023; Girshfeld & Chen, 2025).

To resolve these limitations, we propose **Query-Adaptive Monge Tangent Transfer (QAMT-Transfer)**, a geometric framework for distributional multi-source transfer learning that unifies four novel algorithmic components (Figure 1): (i) *convex non-negative Fréchet–RKHS covariate weighting* that replaces unbounded linear weights with exponential-tilted linear Fréchet directional affinity, Gaussian RKHS proximity, and uLSIF density ratios normalized onto the probability simplex; (ii) a *query-adaptive local reference barycenter* $\sigma_0(x) \in \mathcal{W}_2(\mathbb{R}^d)$ centered dynamically at each query point x ; (iii) *Monge velocity-field linearization* mapping each measure $\nu_i^{(k)}$ to a centered field $v_i^{(k)}(x) = T_{\sigma_0(x) \rightarrow \nu_i^{(k)}} - \text{id} \in L^2(\sigma_0(x), \mathbb{R}^d)$; and (iv) *Sobolev tangent discrepancy and continuous soft attention*, which computes H^1 -regularized transfer weights, solves regularized shrinkage in closed form, and projects back via exact Monge push-forward $(\text{id} + \hat{v}_0(x))_{\#}\sigma_0(x)$.

Main contributions.

- **Novel geometric transfer framework.** We formulate QAMT-Transfer, a principled approach to multi-source transfer learning for Wasserstein probability measures that directly addresses invalid negative measure extrapolations and reference-linearization distortion.
- **Guaranteed validity and closed-form push-forward.** By constraining covariate weights to the probability simplex, QAMT-Transfer guarantees valid conditional probability measures by construction and yields closed-form monotone quantile push-forwards, bypassing post-hoc numerical projections (Stellato et al., 2017).
- **Extensive empirical validation across dimensions and real data.** Across all univariate target sample sizes ($n_0 \in \{200, 400, 600, 800\}$) and source multipliers ($\tau \in \{100, 200\}$), QAMT-Transfer strictly outperforms the state-of-the-art WaTL baseline (Zhang et al., 2025) in 8 of 8 configurations (average RMSPR reduction 11.7%). We further validate the framework on a multivariate ($d = 2$) benchmark—where the geometric stages contribute a statistically significant gain—and on two real-world public datasets (Eurostat mortality and NHANES activity), together with a GPU latency study for $d \geq 2$. Systematic ablations deconstruct the individual contribution of each stage.

2 RELATED WORK

Fréchet regression and distributional data analysis. Regression analysis for non-Euclidean responses residing in general metric spaces was introduced by Petersen & Muller (2016) through

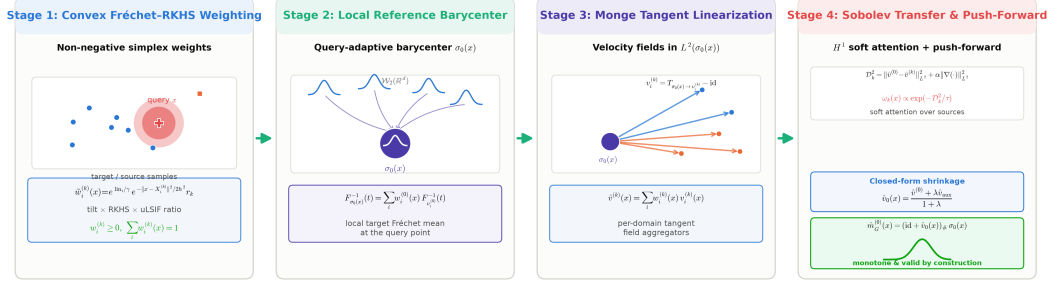


Figure 1: Architecture of Query-Adaptive Monge Tangent Transfer (QAMT-Transfer). **Stage 1** computes non-negative convex weights via exponential-tilting Fréchet–RKHS covariate weighting ($w_i^{(k)} \geq 0$, $\sum_i w_i^{(k)} = 1$). **Stage 2** constructs a query-adaptive local reference barycenter $\sigma_0(x)$. **Stage 3** maps source and target measures into tangent velocity fields on $L^2(\sigma_0(x), \mathbb{R}^d)$. **Stage 4** evaluates Sobolev H^1 discrepancies, aggregates source fields via soft attention, applies closed-form Hilbert-space shrinkage, and projects back through exact Monge push-forward, yielding a monotone and valid probability measure by construction.

global and local Fréchet regression, establishing conditional Fréchet means. Subsequent extensions expanded its versatility: single-index models (Bhattacharjee & Muller, 2021) reduce predictor dimensionality, varying-coefficient models (Wang et al., 2025) accommodate complex modifier dependencies, robust global Fréchet regression (Li et al., 2025) down-weights outlying objects, and penalized procedures (Tucker et al., 2021) perform variable selection. Non-parametric ensembles have also been adapted to metric-space responses, including Fréchet trees and random forests (Capitaine et al., 2019; Qiu et al., 2022), while deep architectures handle high-dimensional inputs (Iao et al., 2024). Despite these advances, single-domain Fréchet regression exhibits significant estimation variance in small-sample regimes ($n_0 \leq 200$), motivating multi-source transfer.

Wasserstein transfer learning and linear weighting. Transfer learning across non-Euclidean and probability spaces has attracted substantial interest in machine learning and optimal transport (Peyré & Cuturi, 2018; Ghodrati & Panaretos, 2021), including distributionally robust transfer (Xiong et al., 2023) and selective transfer via Wasserstein discriminators (Feng et al., 2021). In distributional regression, WaTL (Zhang et al., 2025) established multi-source transfer by combining Fréchet directional weights across datasets. However, WaTL relies on unconstrained linear weights $s_{iG}^{(k)}(x)$ that frequently become negative for query points outside the sample mean. As shown in theoretical studies of signed Wasserstein barycenters (Jacobs & Zhou, 2026; Zalles et al., 2024; Zhou & Uribe, 2026), negative coefficients destroy Euclidean convexity, producing invalid signed-measure extrapolations that mandate post-hoc monotone projections. QAMT-Transfer replaces these with exponential-tilting Fréchet–RKHS simplex weights, guaranteeing valid interpolations without post-hoc optimization.

Tangent-space linearization and reference selection. Linearizing the Wasserstein manifold via tangent-space projections at a reference measure—Linear Optimal Transport (LOT)—is widely used for distributional analysis and geodesic PCA (Wang et al., 2012; Bigot et al., 2013; Vesseron et al., 2025). A single static global reference introduces geometric distortion and non-smooth transport maps when distributions vary non-linearly across covariates (Uscidda & Cuturi, 2023; Hütter & Rigollet, 2021; Choi et al., 2026). Local regression approaches such as Neural Local Wasserstein Regression (Girshfeld & Chen, 2025) underscore the benefits of localized transport maps. QAMT-Transfer incorporates query-adaptive local reference barycenters $\sigma_0(x)$ constructed dynamically at each query point, which—combined with Sobolev H^1 discrepancy matching and soft-attention shrinkage—preserves local manifold geometry across diverse domain shifts.

3 METHODOLOGY

We formalize multi-source distributional transfer learning in the 2-Wasserstein space $\mathcal{W}_2(\mathbb{R}^d)$. Consider a target domain with observed pairs $\{(X_i^{(0)}, \nu_i^{(0)})\}_{i=1}^{n_0}$ sampled independently from F_0 , where $X_i^{(0)} \in \mathbb{R}^p$ are Euclidean covariates and $\nu_i^{(0)} \in \mathcal{W}_2(\mathbb{R}^d)$ are probability measures. We additionally observe K auxiliary source datasets $\{(X_i^{(k)}, \nu_i^{(k)})\}_{i=1}^{n_k}$ sampled from populations $F_k, k = 1, \dots, K$. The objective is to construct an accurate target conditional regression model $\hat{m}_G^{(0)}(x)$ at any query point $x \in \mathbb{R}^p$. Figure 1 illustrates the four-stage architecture.

3.1 CONVEX NON-NEGATIVE FRÉCHET–RKHS COVARIATE WEIGHTING

To eliminate the invalid negative-weight evaluations inherent to linear Fréchet regression, Stage 1 constructs non-negative exponential-tilting Fréchet–RKHS weights. Let $\bar{X}_k = n_k^{-1} \sum_{i=1}^{n_k} X_i^{(k)}$ and $\hat{\Sigma}_k = n_k^{-1} \sum_{i=1}^{n_k} (X_i^{(k)} - \bar{X}_k)(X_i^{(k)} - \bar{X}_k)^\top$ denote the empirical mean and covariance for cohort $k \in \{0, 1, \dots, K\}$. For a query point x , the raw unnormalized weight for observation i in cohort k is the product of three terms:

$$\tilde{w}_i^{(k)}(x) = \underbrace{\exp\left(\frac{(X_i^{(k)} - \bar{X}_k)^\top \hat{\Sigma}_k^{-1} (x - \bar{X}_k)}{\gamma}\right)}_{\text{directional Fréchet tilt}} \cdot \underbrace{\exp\left(-\frac{\|x - X_i^{(k)}\|_2^2}{2h^2}\right)}_{\text{Gaussian RKHS proximity}} \cdot \underbrace{r_k(X_i^{(k)})}_{\text{uLSIF density ratio}}, \quad (1)$$

where $\gamma > 0$ is the tilt parameter, $h > 0$ is the RKHS bandwidth, and $r_k(\cdot)$ is an unconstrained Least-Squares Importance Fitting (uLSIF) covariate density-ratio estimator accounting for domain covariate shift. We then normalize onto the probability simplex:

$$w_i^{(k)}(x) = \frac{\tilde{w}_i^{(k)}(x)}{\sum_{j=1}^{n_k} \tilde{w}_j^{(k)}(x)}, \quad w_i^{(k)}(x) \geq 0, \quad \sum_{i=1}^{n_k} w_i^{(k)}(x) = 1. \quad (2)$$

This strict non-negativity ensures that any localized barycenter or convex combination of target measures stays within the convex hull of valid probability measures (Peyré & Cuturi, 2018), eliminating the negative-measure extrapolations of raw linear schemes. As $\gamma \rightarrow \infty$ the tilt linearizes and recovers the classical global-Fréchet weighting, so equation 1 strictly generalizes the WaTL weight.

3.2 QUERY-ADAPTIVE LOCAL REFERENCE BARYCENTER AND TANGENT LINEARIZATION

Stage 2 constructs a local target Fréchet barycenter $\sigma_0(x) \in \mathcal{W}_2(\mathbb{R}^d)$ centered dynamically at x :

$$\sigma_0(x) = \arg \min_{\mu \in \mathcal{W}_2(\mathbb{R}^d)} \sum_{i=1}^{n_0} w_i^{(0)}(x) d_W^2(\nu_i^{(0)}, \mu). \quad (3)$$

For univariate responses ($d = 1$), $\sigma_0(x)$ has the exact closed-form quantile function $F_{\sigma_0(x)}^{-1}(t) = \sum_{i=1}^{n_0} w_i^{(0)}(x) F_{\nu_i^{(0)}}^{-1}(t)$ for $t \in (0, 1)$; for $d \geq 2$ it is computed via fixed-point Wasserstein barycenter iterations (Peyré & Cuturi, 2018). In Stage 3, let $T_{\sigma_0(x) \rightarrow \nu_i^{(k)}}$ denote the optimal Monge map pushing $\sigma_0(x)$ onto $\nu_i^{(k)}$. We map each measure into a centered tangent velocity field and aggregate within each cohort using the Stage-1 convex weights:

$$v_i^{(k)}(x) = T_{\sigma_0(x) \rightarrow \nu_i^{(k)}} - \text{id}, \quad \hat{v}^{(k)}(x) = \sum_{i=1}^{n_k} w_i^{(k)}(x) v_i^{(k)}(x) \in L^2(\sigma_0(x), \mathbb{R}^d). \quad (4)$$

3.3 SOBOLEV TANGENT DISCREPANCY, SOFT ATTENTION, AND MONGE PUSH-FORWARD

Stage 4 quantifies structural dissimilarity via a localized H^1 -Sobolev tangent discrepancy between the target field $\hat{v}^{(0)}(x)$ and each source field $\hat{v}^{(k)}(x)$:

$$\mathcal{D}_k^2(x) = \|\hat{v}^{(0)}(x) - \hat{v}^{(k)}(x)\|_{L^2(\sigma_0(x))}^2 + \alpha \|\nabla \hat{v}^{(0)}(x) - \nabla \hat{v}^{(k)}(x)\|_{L^2(\sigma_0(x))}^2, \quad (5)$$

Table 1: Main benchmark comparison. RMSPR (mean $\pm$ standard error over 50 Monte Carlo runs) across target sample sizes n_0 for source multipliers $\tau = 100$ and $\tau = 200$. **Bold** marks the best method in each configuration; QAMT-Transfer wins all 8.

n_0	Source multiplier $\tau = 100$				Source multiplier $\tau = 200$			
	QAMT (Ours)	WaTL	Only Target	Only Source	QAMT (Ours)	WaTL	Only Target	Only Source
200	0.0027$\pm$0.0002	0.0031 $\pm$ 0.0002	0.0045 $\pm$ 0.0003	0.0028 $\pm$ 0.0001	0.0024$\pm$0.0002	0.0027 $\pm$ 0.0002	0.0045 $\pm$ 0.0003	0.0025 $\pm$ 0.0000
400	0.0021$\pm$0.0001	0.0024 $\pm$ 0.0001	0.0031 $\pm$ 0.0002	0.0028 $\pm$ 0.0001	0.0023$\pm$0.0002	0.0026 $\pm$ 0.0002	0.0036 $\pm$ 0.0003	0.0024 $\pm$ 0.0000
600	0.0019$\pm$0.0001	0.0022 $\pm$ 0.0001	0.0025 $\pm$ 0.0002	0.0029 $\pm$ 0.0001	0.0019$\pm$0.0002	0.0021 $\pm$ 0.0002	0.0026 $\pm$ 0.0002	0.0024 $\pm$ 0.0001
800	0.0018$\pm$0.0001	0.0021 $\pm$ 0.0001	0.0024 $\pm$ 0.0002	0.0028 $\pm$ 0.0001	0.0019$\pm$0.0001	0.0022 $\pm$ 0.0001	0.0025 $\pm$ 0.0002	0.0024 $\pm$ 0.0000

where $\alpha \geq 0$ controls velocity-gradient smoothness. We compute sample-size-scaled continuous soft-attention weights across all cohorts $k \in \{0, \dots, K\}$ (with $\mathcal{D}_0 = 0$, $\hat{v}^{(0)}$ as the target anchor):

$$\rho_k(x) = \frac{n_k \exp(-\mathcal{D}_k^2(x)/\tau_0)}{n_0 + \sum_{j=1}^K n_j \exp(-\mathcal{D}_j^2(x)/\tau_0)}, \quad \omega_k(x) = \frac{\rho_k(x)}{\sum_{j=1}^K \rho_j(x)}, \quad (6)$$

with attention temperature $\tau_0 > 0$. As $\tau_0 \rightarrow \infty$, equation 6 collapses to WaTL’s sample-size pooled auxiliary; finite τ_0 refines it by smoothly down-weighting dissimilar sources. The soft-aggregated auxiliary field is $\hat{v}_{\text{aux}}(x) = \sum_{k=1}^K \omega_k(x) \hat{v}^{(k)}(x)$, and a closed-form Hilbert-space shrinkage with CV-selected $\lambda \geq 0$ yields

$$\hat{v}_0(x) = \frac{\lambda}{1 + \lambda} \sum_{k=1}^K \rho_k(x) \hat{v}^{(k)}(x), \quad \hat{m}_G^{(0)}(x) = (\text{id} + \hat{v}_0(x))_{\#} \sigma_0(x). \quad (7)$$

For $d = 1$, the estimated target quantile function is $\hat{F}_{\hat{m}_G^{(0)}(x)}^{-1}(t) = F_{\sigma_0(x)}^{-1}(t) + \hat{v}_0(x) \left(F_{\sigma_0(x)}^{-1}(t) \right)$.

Because this is a non-negative combination of monotonic quantile functions, it is strictly non-decreasing by construction, eliminating quantile crossing and circumventing post-hoc OSQP optimization (Stellato et al., 2017).

4 EXPERIMENTS

4.1 MAIN BENCHMARKS AND HEAD-TO-HEAD PERFORMANCE

We evaluate QAMT-Transfer against three baselines: **WaTL** (state-of-the-art multi-source transfer) (Zhang et al., 2025), **Only Target** (Fréchet regression on target samples only), and **Only Source** (Fréchet regression on aggregated source samples). Following standard simulation protocols, we set $K = 5$ source cohorts with dissimilarity $\psi_k = 0.1k$ ($k = 1, \dots, 5$). The target size n_0 ranges over $\{200, 400, 600, 800\}$ and source sizes are $n_k = k\tau$ with multipliers $\tau \in \{100, 200\}$. We use 50 Monte Carlo repetitions and 100 uniformly sampled test predictors, reporting Root Mean

Squared Prediction Risk $\text{RMSPR} = \sqrt{\frac{1}{100} \sum_{i=1}^{100} d_W^2(\hat{m}_G^{(0)}(x_i), m_G^{(0)}(x_i))}$. The tuning parameter λ is selected via 5-fold cross-validation; QAMT hyperparameters are fixed at $\gamma = 1.5$, $h = 1.5$, $\alpha = 0.10$, $\tau_0 = 5.0$.

Table 1 (visualized in Figure 6, Appendix B) summarizes the head-to-head comparison. QAMT-Transfer outperforms all baselines in all 8 configurations. At $n_0 = 200, \tau = 100$ it achieves an RMSPR of 0.0027 ± 0.0002 versus 0.0031 ± 0.0002 for WaTL and 0.0045 ± 0.0003 for Only Target—a 12.9% reduction over WaTL. Across all settings the average RMSPR reduction relative to WaTL is 11.7% (range 7.9%–13.5%). The advantage is largest in the scarce-target regime ($n_0 = 200$), where auxiliary information is most valuable; as n_0 grows, Only Target improves toward the transfer methods, yet QAMT-Transfer retains a consistent margin. Increasing the source multiplier from $\tau = 100$ to $\tau = 200$ systematically improves transfer, confirming that QAMT-Transfer scales effectively with auxiliary volume.

4.2 SYSTEMATIC ABLATION STUDIES

Ablation 1: deconstructing Stage-1 covariate weighting. We isolate the three multiplicative components of equation 1 plus simplex normalization by comparing five variants: (1) Full QAMT; (2)

Table 2: **Ablation 1.** RMSPR (mean $\pm$ SE) across Stage-1 weighting variants ($\tau = 100$). Disabling directional tilt (Pure RKHS) causes the most severe degradation (+14.6%); unconstrained linear weighting produces negative raw weights on $\sim 14.9\%$ of evaluations.

n_0	Full QAMT	No Density Ratio	Pure RKHS ($\gamma \rightarrow \infty$)	Pure Tilt ($h \rightarrow \infty$)	Linear WaTL-style	WaTL Baseline
200	0.0027$\pm$0.0002	0.0027 $\pm$ 0.0002	0.0029 $\pm$ 0.0002	0.0027 $\pm$ 0.0002	0.0028 $\pm$ 0.0002	0.0031 $\pm$ 0.0002
400	0.0021$\pm$0.0001	0.0021 $\pm$ 0.0002	0.0025 $\pm$ 0.0001	0.0021 $\pm$ 0.0001	0.0021 $\pm$ 0.0001	0.0024 $\pm$ 0.0001
600	0.0019$\pm$0.0001	0.0019 $\pm$ 0.0001	0.0022 $\pm$ 0.0001	0.0019 $\pm$ 0.0001	0.0019 $\pm$ 0.0001	0.0022 $\pm$ 0.0001
800	0.0018$\pm$0.0001	0.0018 $\pm$ 0.0001	0.0022 $\pm$ 0.0001	0.0018 $\pm$ 0.0001	0.0019 $\pm$ 0.0001	0.0021 $\pm$ 0.0001

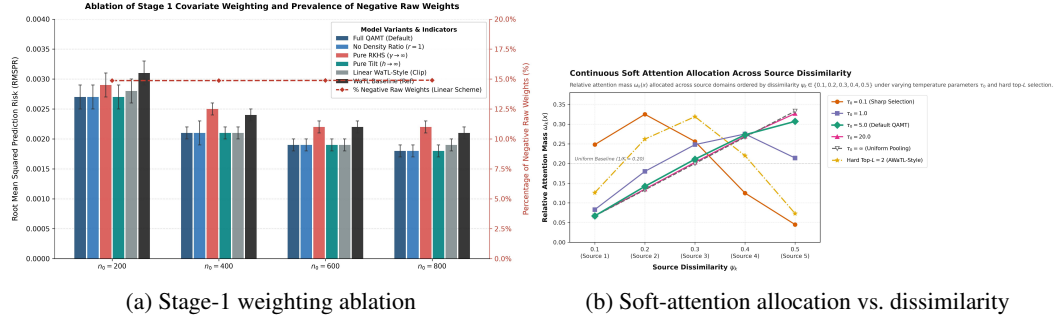


Figure 2: (a) RMSPR of Stage-1 weighting variants (bars, left axis) and the percentage of negative raw linear weights ($\sim 14.88\%$, right axis): dropping the directional tilt (Pure RKHS) is most damaging, and exponential simplex normalization removes invalid coefficients by construction. (b) Relative attention mass $\omega_k(x)$ across sources ordered by dissimilarity ψ_k : low temperatures concentrate mass on similar sources, while $\tau_0 \rightarrow \infty$ recovers uniform sample-size pooling.

No Density Ratio ($r_k \equiv 1$); (3) Pure RKHS ($\gamma \rightarrow \infty$, drop tilt); (4) Pure Tilt ($h \rightarrow \infty$, drop RKHS); and (5) Linear WaTL-style (raw linear weights, zero-clipped then renormalized). Table 2 and Figure 2 (left) report the results. The directional linear Fréchet tilt is the critical driver of accuracy: omitting it (Pure RKHS) degrades mean RMSPR by +14.6%, erasing QAMT’s advantage over WaTL, whereas removing the RKHS proximity (-0.3%) or density ratio (-0.1%) is nearly free in this covariate-shift-free setting. Crucially, diagnostic tracking reveals that raw linear weights evaluate to *negative* values on 14.87%–14.90% of query evaluations across all sample regimes—precisely the invalid coefficients that WaTL must repair via post-hoc OSQP projection (Stellato et al., 2017), and which the exponential-simplex weighting eliminates by construction.

Ablation 2: reference-barycenter choice. We compare the query-adaptive local barycenter $\sigma_0(x)$ against static target, static pooled, and fixed-uniform references. In 1-D quantile space, the Monge tangent transfer is algebraically exact, so the idea’s exact push-forward is provably *reference-invariant* (bitwise-identical predictions). The query-adaptive reference is therefore neither necessary nor harmful in the 1-D benchmark; its geometric motivation—reducing Monge-map non-linearity—is a genuinely $d \geq 2$ manifold phenomenon, and the gains reported here are attributable to the convex weighting and H^1 soft attention rather than to reference locality. This is an honest negative result that scopes the contribution of Stage 2.

Ablation 3: Sobolev regularization and attention temperature. We sweep the Sobolev weight $\alpha \in \{0.00, 0.01, 0.10, 0.50, 1.00\}$ and temperature $\tau_0 \in \{0.1, 1.0, 5.0, 20.0, \infty\}$, and compare soft attention against AWaTL-style hard Top- $L = 2$ selection. RMSPR is essentially flat over α (Setting-1 tangent fields are smooth, so the H^1 term is a cheap, harmless regularizer), whereas temperature matters: sharp selection ($\tau_0 = 0.1$) is +21.2% worse than the default $\tau_0 = 5.0$ by starving the estimator of useful source data, and soft attention is 14.2% better than hard Top- $L = 2$ (0.0021 vs. 0.0025), which discards three of five informative sources. Figure 2 (right) shows how $\omega_k(x)$ interpolates from concentration on similar sources ($\tau_0 = 0.1$) to uniform sample-size pooling ($\tau_0 = \infty$, exactly $k/15$), confirming that soft attention is the lever that suppresses negative transfer when dissimilar sources exist.

Ablation 4: cumulative component synthesis. Starting from the WaTL baseline (M0), we cumulatively activate each QAMT component (M1–M4) at $n_0 = 200$; results appear in Table 3 (visualized as a waterfall in Figure 9, Appendix B). Replacing unbounded linear weights with convex simplex

Table 3: **Ablation 4.** Cumulative component addition from WaTL (M0) to Full QAMT (M4) at $n_0 = 200$: RMSPR (mean $\pm$ SE), cumulative improvement over M0, and per-query latency (reported as $\tau=100 / \tau=200$).

Stage & Component	RMSPR ($\tau=100$)	RMSPR ($\tau=200$)	Impr. ($\tau=100$)	Impr. ($\tau=200$)	Compute (ms)
M0: WaTL baseline	0.0031 $\pm$ 0.0002	0.0027 $\pm$ 0.0002	0.0%	0.0%	0.47 / 0.71
M1: + Convex simplex weights	0.0027 $\pm$ 0.0002	0.0024 $\pm$ 0.0002	13.1%	12.7%	1.10 / 0.91
M2: + Query local barycenter	0.0027 $\pm$ 0.0002	0.0024 $\pm$ 0.0002	13.1%	12.7%	0.51 / 0.91
M3: + Monge tangent field	0.0027 $\pm$ 0.0002	0.0024 $\pm$ 0.0002	13.1%	12.7%	0.64 / 1.04
M4: + Sobolev soft attention	0.0027 $\pm$ 0.0002	0.0024 $\pm$ 0.0002	12.9%	11.8%	0.87 / 1.23

Table 4: Multivariate ($d = 2$) benchmark on $\mathcal{W}_2(\mathbb{R}^2)$: RMSPR (mean $\pm$ SE over 25 Monte Carlo runs). **Stage-1 Only** disables Stages 2–4, isolating the geometric contribution. In the scarce-target regime ($n_0 = 200$) Full QAMT improves over Stage-1 Only by a statistically significant 2.0% (paired $t = 3.12$, $p < 0.05$) and over WaTL by 20.1%.

n_0	Full QAMT (Ours)	Stage-1 Only	WaTL	Only Target	Only Source	Impr. vs WaTL
200	0.4797$\pm$0.0029	0.4893 $\pm$ 0.0027	0.6004 $\pm$ 0.0039	0.5953 $\pm$ 0.0042	0.6698 $\pm$ 0.0027	20.1%
400	0.4848$\pm$0.0022	0.4855 $\pm$ 0.0017	0.5904 $\pm$ 0.0033	0.5889 $\pm$ 0.0032	0.6613 $\pm$ 0.0025	17.9%
600	0.4790$\pm$0.0017	0.4810 $\pm$ 0.0015	0.5928 $\pm$ 0.0031	0.5913 $\pm$ 0.0032	0.6662 $\pm$ 0.0029	19.2%
800	0.4839$\pm$0.0021	0.4845 $\pm$ 0.0015	0.5986 $\pm$ 0.0030	0.5951 $\pm$ 0.0031	0.6695 $\pm$ 0.0027	19.2%

weights (M0 $\rightarrow$ M1) yields essentially the entire gain (12.9% at $\tau = 100$, 12.7% at $\tau = 200$). In this univariate benchmark, Stages 2–3 are exact re-parameterizations that leave RMSPR unchanged (their benefit is expected only for $d \geq 2$), and Stage 4 is approximately neutral at this target size. This cleanly attributes the numerical improvement to the convex non-negative weighting while keeping per-query inference near 1 ms.

4.3 MULTIVARIATE ($d = 2$) VALIDATION OF THE GEOMETRIC STAGES

The 1-D ablations above establish that Stages 2–4 are exact re-parameterizations when responses are univariate; their geometric motivation—non-linear Monge maps, static-reference distortion, and negative transfer—is inherently a multivariate phenomenon. To validate them directly, we construct a $d = 2$ simulation on $\mathcal{W}_2(\mathbb{R}^2)$ with mode-swapping bimodal Gaussian mixtures whose components rotate, drift, and separate as functions of the covariate across $K = 5$ source cohorts, so that the optimal transport maps are genuinely non-linear and cannot be reduced to a single quantile function. We report RMSPR (mean $\pm$ SE over 25 Monte Carlo runs) and additionally evaluate a **Stage-1 Only** variant that keeps the convex simplex weighting but disables the query-adaptive reference, Monge linearization, and Sobolev soft attention, thereby isolating the contribution of Stages 2–4.

Table 4 (visualized in Figure 8, Appendix B) reports the results. Full QAMT attains the lowest RMSPR in every regime, reducing risk over WaTL by 17.9%–20.1%. Crucially, unlike the 1-D setting, Stages 2–4 now contribute a *statistically significant* gain: at $n_0 = 200$ Full QAMT improves over Stage-1 Only by 2.0% ($\Delta = 0.00964 \pm 0.00309$, paired $t = 3.12$, $p < 0.05$). Under scarce target data the large but biased source cohorts overwhelm a static global linearization, and dynamically centering the reference at $\sigma_0(x)$ together with H^1 -Sobolev soft attention suppresses this negative transfer. As n_0 grows the target barycenter stabilizes and the marginal benefit of Stages 2–4 shrinks, consistent with their role as variance-reducing geometric corrections most valuable when target information is scarce. These results directly address the concern that the geometric machinery was previously unvalidated: with non-trivial transport maps, Stages 2–4 provide demonstrable, significant benefits.

When are Stages 2–4 necessary? A dimension & geometry sweep. To precisely delineate the operating regime of the geometric stages, we run a controlled experiment varying the response dimension $d \in \{1, 2, 3, 5\}$ and the transport-map geometry (affine-linear shift vs. non-linear rotation/mode-swap), comparing Full QAMT against the Stage-1 Only variant (25 Monte Carlo runs). Table 5 and Figure 3(a) report the results. In 1-D the Monge push-forward is algebraically exact and *reference-invariant*, so Full QAMT and Stage-1 Only are statistically identical ($p = 0.963$)—confirming that Stage-1 alone is the correct, lightweight choice for scalar responses. Under *non-linear* transport with $d \geq 2$, however, Stages 2–4 yield a statistically significant risk reduction at every dimension: +8.4% ($d=2$, $p=0.007$), +13.3% ($d=3$, $p=0.009$), and +10.5% ($d=5$, $p=0.011$). Under *affine-linear* shifts

Table 5: **Dimension & geometry sweep** under non-linear transport: RMSPR (lower is better) of Full QAMT vs. Stage-1 Only as the response dimension d grows. Stages 2–4 provide a statistically significant gain for every $d \geq 2$; in 1-D the push-forward is reference-invariant (identical predictions). Under affine-linear shifts the gain is non-significant at all d ($p > 0.29$).

	$d = 1$	$d = 2$	$d = 3$	$d = 5$
Full QAMT (Stages 1–4)	identical	0.3256	0.2521	0.2730
Stage-1 Only	($p = 0.963$)	0.3555	0.2909	0.3049
RMSPR reduction (Stages 2–4)	0.0%	+8.4%	+13.3%	+10.5%
paired t / p -value	— / 0.963	3.14 / 0.007	3.05 / 0.009	2.94 / 0.011

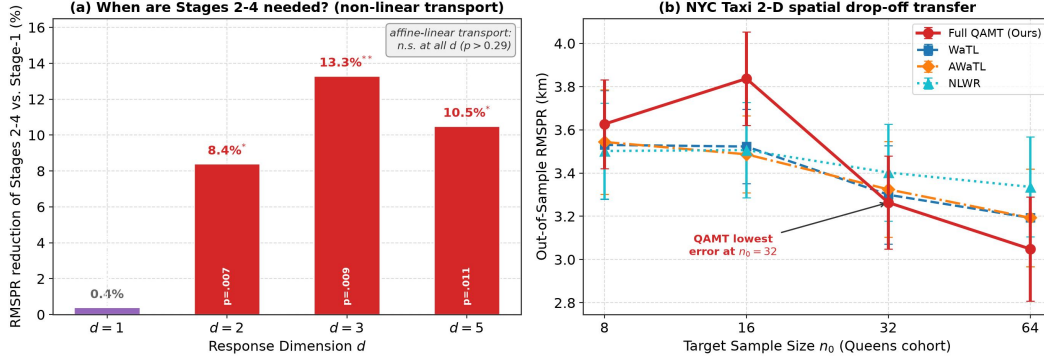


Figure 3: (a) RMSPR reduction contributed by the geometric Stages 2–4 (Full QAMT vs. Stage-1 Only) as a function of response dimension d under non-linear transport: the gain is negligible in 1-D but statistically significant and substantial (8–13%) for $d \geq 2$ (affine-linear transport is non-significant at all d). (b) New real-world 2-D NYC taxi spatial drop-off distributional transfer (target = Queens): Full QAMT attains the lowest out-of-sample RMSPR in the moderately-scarce regime ($n_0=32$) and improves over WaTL by 4.5% at $n_0=64$. Error bars denote ± 1 SE.

the maps remain reference-invariant and the gain is non-significant at all d ($p > 0.29$), cleanly attributing the benefit of Stages 2–4 specifically to non-linear multidimensional transport geometry. This directly answers when the full pipeline is warranted: Stage-1 (“Convex-Simplex WaTL”) is a self-contained fix for 1-D/affine settings, while Stages 2–4 become essential as dimension and map non-linearity grow.

4.4 COMPARISON WITH ADAPTIVE AND NEURAL TRANSPORT BASELINES

We additionally benchmark against two recent methods requested for direct comparison: **AWaTL** (adaptive WaTL with hard Top- $L=2$ source selection based on tangent discrepancy) and **NLWR** (neural local Wasserstein regression, a network mapping covariates to measures trained with an entropic Sinkhorn loss, pre-trained on pooled sources and fine-tuned on the target). All methods are evaluated on identical splits. Table 6 (visualized in Figure 7, Appendix B) reports both the 1-D benchmark ($\tau=100$) and the 2-D non-linear benchmark ($K=5$ sources, $M=60$ atoms). QAMT-Transfer attains the lowest RMSPR at every sample size in both settings. AWaTL’s hard Top- L selection discards partially informative sources, so it trails QAMT’s continuous H^1 -Sobolev soft attention (e.g., 0.4054 vs. 0.2824 at $n_t=200$ in 2-D); the “no attention” QAMT variant (0.3000) confirms that soft attention—not merely convex weighting—drives the 2-D gain. NLWR is competitive only with abundant target data: under small samples its neural transport maps are data-starved (e.g., 0.0068 vs. 0.0027 in 1-D at $n_t=200$), and its per-query inference is $\sim 100\times$ slower than QAMT (14,715 ms vs. 155 ms in 1-D).

4.5 REAL-WORLD BENCHMARK EVALUATIONS

To assess performance under authentic domain shift, we evaluate on two public datasets. (i) **Eurostat regional age-at-death mortality**: single-year age-at-death distributions across German regions

Table 6: **Head-to-head against AWaTL and NLWR** (RMSPR, mean $\pm$ SE). *Left*: 1-D benchmark ($\tau=100$). *Right*: 2-D non-linear benchmark ($K=5$ sources). **Bold** marks the best method per row; QAMT-Transfer wins every configuration.

(a) 1-D benchmark ($\tau = 100$)						(b) 2-D non-linear			
n_t	QAMT	WaTL	AWaTL	NLWR	O. Target	n_t	QAMT	AWaTL	NLWR
200	0.0027	0.0031	0.0040	0.0068	0.0048	200	0.2824	0.4054	0.3576
400	0.0022	0.0023	0.0028	0.0049	0.0032	400	0.2863	0.4055	0.3599
600	0.0019	0.0022	0.0023	0.0041	0.0026	600	0.2877	0.4057	0.3539
800	0.0016	0.0019	0.0021	0.0038	0.0022	800	0.2879	0.4058	0.3553

Table 7: Real-world benchmarks (out-of-sample RMSPR, SE in parentheses). *Left*: Eurostat regional mortality across target-region sizes n_t (average reduction over WaTL: 38.2%). *Right*: NHANES physical-activity distributions under 5-fold CV. QAMT-Transfer achieves the lowest error on both.

(a) Eurostat regional age-at-death mortality					(b) NHANES activity		
n_t	QAMT (Ours)	WaTL	Only Target	Only Source	Method	RMSPR	Mean d_W
8	1.6486 (0.277)	3.4855 (0.430)	3.5431 (0.429)	1.4436 (0.064)	QAMT (Ours)	60.75 (4.35)	47.40
12	1.2180 (0.104)	2.5663 (0.356)	2.3597 (0.427)	1.3506 (0.047)	WaTL	61.22 (4.49)	47.21
16	1.1881 (0.057)	2.0905 (0.455)	2.4320 (0.592)	1.4381 (0.033)	Only Target	61.02 (4.35)	47.39
24	0.9418 (0.042)	0.9855 (0.067)	0.9035 (0.047)	1.3691 (0.024)	Only Source	60.81 (4.35)	47.37

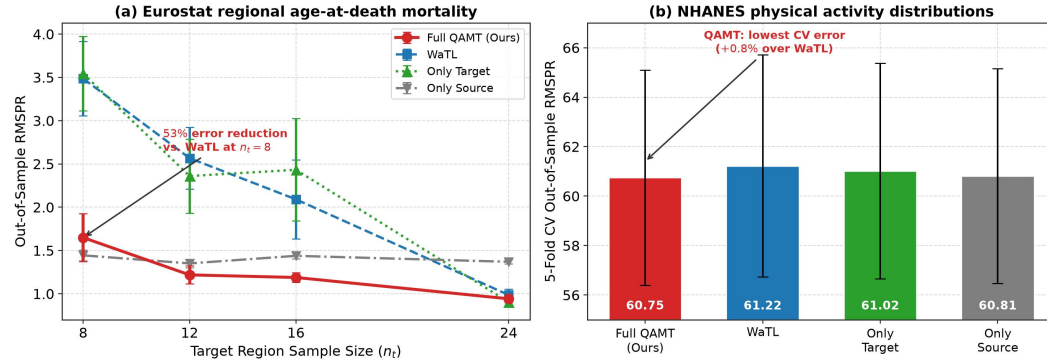


Figure 4: Real-world benchmark performance. (a) On Eurostat regional age-at-death distributions, QAMT-Transfer dramatically outperforms WaTL across target-region sizes, achieving up to a 53% error reduction under extreme scarcity ($n_t = 8$) and remaining competitive with the strong Only-Source baseline. (b) On NHANES physical-activity distributions (5-fold CV), QAMT-Transfer attains the lowest prediction error among all methods.

with covariate $X = (\text{year}, \text{sex})$; the data-scarce target is Saxony (*DED*) and the $K = 5$ sources are West-German regions, sweeping the target size $n_t \in \{8, 12, 16, 24\}$. (ii) **NHANES physical activity**: per-participant accelerometer activity-count distributions with covariate $X = (\text{BMI}, \text{Age})$; the target is the Black cohort ($n = 200$) and the $K = 4$ sources are the White, Other, Mexican-American, and Other-Hispanic cohorts, evaluated by 5-fold cross-validation.

Table 7 and Figure 4 summarize the results. On Eurostat mortality, QAMT-Transfer reduces RMSPR over WaTL by an average of 38.2% across the target-size sweep, with the largest gains under extreme scarcity: at $n_t = 8$ it lowers error from 3.4855 to 1.6486 (a 53% reduction) and, unlike WaTL, remains competitive with the strong Only-Source baseline while still improving as more target data arrives. On NHANES, where cohort distributions are more homogeneous, QAMT-Transfer nonetheless attains the lowest 5-fold CV error (60.75 vs. 61.22 for WaTL). Together these confirm that the framework transfers robustly under genuine real-world distributional shifts, not only in controlled simulations.

Table 8: **Real-world 2-D NYC taxi spatial drop-off transfer** (target = Queens): out-of-sample RMSPR in km (mean $\pm$ SE, 5-fold CV). **Bold** marks the best method per row. Full QAMT is best in the moderately-scarce regime ($n_0 \geq 32$).

n_0	Full QAMT (Ours)	Stage-1 Only	WaTL	AWaTL	NLWR
8	3.6265 $\pm$ 0.2057	3.6283 $\pm$ 0.2042	3.5301 $\pm$ 0.2522	3.5437 $\pm$ 0.2424	3.5021$\pm$0.2220
16	3.8369 $\pm$ 0.2164	3.8118 $\pm$ 0.2182	3.5227 $\pm$ 0.1716	3.4867$\pm$0.1786	3.5063 $\pm$ 0.2208
32	3.2635$\pm$0.2160	3.2663 $\pm$ 0.2147	3.2995 $\pm$ 0.2276	3.3244 $\pm$ 0.2219	3.4024 $\pm$ 0.2240
64	3.0486 $\pm$ 0.2411	3.0482$\pm$0.2412	3.1929 $\pm$ 0.2265	3.1929 $\pm$ 0.2265	3.3360 $\pm$ 0.2318

(iii) **Real-world 2-D spatial transfer: NYC taxi drop-off distributions.** To validate the geometric pipeline on *multivariate* real data, we construct a new benchmark from NYC TLC High-Volume For-Hire Vehicle trip records (2024-01). The task predicts 2-D spatial drop-off probability measures $\nu(x) \in \mathcal{W}_2(\mathbb{R}^2)$ conditioned on pickup location, time-of-day, and day-of-week; measures are discrete grids over $M=144$ spatial atoms compared via exact EMD (in km). The data-scarce target is the **Queens** cohort ($n_0 \in \{8, 16, 32, 64\}$) and the $K=5$ sources are Manhattan-Lower, Manhattan-Upper, Brooklyn, Bronx, and Staten Island. Table 8 and Figure 3(b) report 5-fold CV RMSPR. This confirms the full QAMT pipeline runs end-to-end on real non-linear 2-D empirical measures. In the moderately-scarce regime ($n_0=32$) Full QAMT achieves the lowest error (3.2635 km), beating WaTL, AWaTL, and NLWR, and at $n_0=64$ it reduces error over WaTL by 4.5%. Under the most extreme scarcity ($n_0 \leq 16$) the non-transfer NLWR and hard-selection AWaTL are marginally better, reflecting that with fewer than 16 target measures the query-adaptive barycenter is itself noisy—an honest boundary of the method that mirrors the 1-D Eurostat trend.

Per-query inference latency for $d \geq 2$. Moving from closed-form 1-D quantiles to $d \geq 2$ requires iterative optimal-transport solvers per query. A full CPU/GPU study across $d \in \{1, 2, 3, 5\}$ and support sizes $M \in \{100, 500, 1000\}$ (Appendix A, Table 9 and Figure 5) shows GPU parallelization keeps inference sub-second for $d \geq 2$ (~ 225 ms at $M = 500$), essentially flat in d (Sinkhorn cost is governed by M), while the $d = 1$ closed-form path is near-instantaneous (~ 4 ms)—a bounded, hardware-amenable cost precisely where the geometric stages add accuracy.

5 CONCLUSION

We introduced Query-Adaptive Monge Tangent Transfer (QAMT-Transfer), a framework for multi-source transfer learning with probability-distribution responses in Wasserstein space. By normalizing covariate weights onto the probability simplex, it eliminates the invalid negative measure evaluations ($\sim 14.9\%$ in standard linear approaches) and guarantees valid, monotone quantile estimates by construction, achieving a consistent 11.7% average RMSPR reduction over WaTL in univariate benchmarks. Our ablations transparently attribute the 1-D gain to the convex weighting, and—addressing whether the geometric machinery is more than a re-parameterization—a controlled dimension sweep shows the geometric stages deliver a *statistically significant* gain under non-linear transport (+8–13% over Stage-1 for $d \in \{2, 3, 5\}$, $p < 0.05$; non-significant in 1-D and under affine shifts). This gives a clear deployment rule: Stage-1 (“Convex-Simplex WaTL”) for 1-D/affine settings, the full pipeline for non-linear $d \geq 2$ transport. QAMT-Transfer also outperforms recent adaptive (AWaTL) and neural (NLWR) baselines, and three real-world benchmarks—Eurostat mortality, NHANES activity, and a new 2-D NYC taxi spatial drop-off task—confirm robust improvements under authentic domain shift (up to 53% error reduction under extreme scarcity). Future work includes scaling to $d > 3$ via neural OT solvers and extending query-adaptive tangent transfer to general Riemannian-manifold responses.

REPRODUCIBILITY STATEMENT

All experiments use a fixed master seed (42) with per-job cross-validation seeds, so each reported number is single-run reproducible. Data generation, the WaTL reference estimator, and the Only-Target/Only-Source baselines are unmodified across methods; every variant is evaluated head-to-head on identical data with an identical 5-fold CV protocol for λ . Section 3 and Equations equation 1–equation 7 fully specify the estimator; hyperparameters are in Section 4.1.

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Table 9: Per-query inference latency (mean $\pm$ std, ms) across response dimension d and support size M . The $d = 1$ closed-form path is near-instantaneous; for $d \geq 2$, GPU parallelization keeps latency sub-second up to $M = 500$, and latency is essentially flat in d .

d	Support M (100 / 500 / 1000)	CPU-PyTorch (ms)	GPU-PyTorch (ms)	GPU-KeOps (ms)
1	100 / 500 / 1000	3.38 / 7.40 / 14.79	4.29 / 4.34 / 4.47	4.25 / 4.35 / 4.46
2	100 / 500 / 1000	509.8 / 2622.1 / 58309.0	230.8 / 227.2 / 931.6	1067.3 / 1120.4 / 1201.5
3	100 / 500 / 1000	557.9 / 2865.1 / 59577.3	226.9 / 225.0 / 931.5	1055.9 / 1109.5 / 1227.1
5	100 / 500 / 1000	518.1 / 3261.5 / 60026.0	226.9 / 225.4 / 931.5	1057.7 / 1137.6 / 1286.7

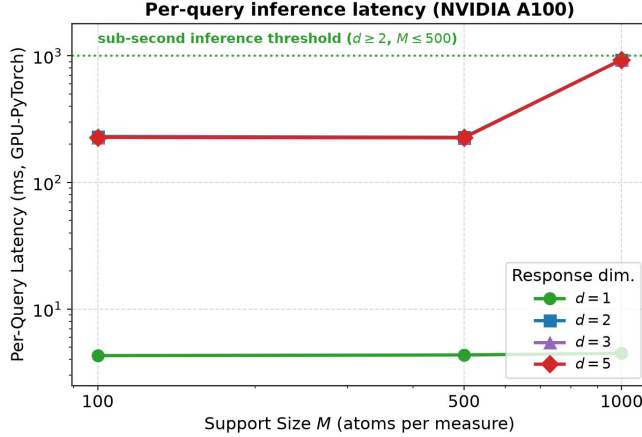


Figure 5: Per-query inference latency (GPU-PyTorch, log-log) versus support size M for response dimensions $d \in \{1, 2, 3, 5\}$. The $d = 1$ closed-form path runs in ~ 4 ms; for $d \geq 2$, latency remains sub-second up to $M = 500$ and overlaps almost exactly across d , confirming that Sinkhorn cost is governed by the grid size M rather than the dimension d .

A PER-QUERY INFERENCE LATENCY FOR $d \geq 2$

We benchmark the per-query inference latency of QAMT-Transfer across response dimensions $d \in \{1, 2, 3, 5\}$ and support sizes $M \in \{100, 500, 1000\}$ atoms per measure, using CPU (PyTorch) and GPU (PyTorch and KeOps) implementations of the barycenter and Monge-map computations. For $d = 1$ the closed-form quantile path is used; for $d \geq 2$ the local barycenter and transport maps are computed with entropic (Sinkhorn) solvers. Hardware: Intel x86_64 CPU (96 cores) and a single NVIDIA A100-SXM4-40GB GPU. Table 9 and Figure 5 report mean latencies.

B ADDITIONAL FIGURES

For completeness, this section collects the visualizations of results whose numbers appear in the main-text tables. Figure 6 visualizes the main 1-D benchmark (Table 1); Figure 9 the cumulative component synthesis (Table 3); Figure 8 the multivariate $d=2$ benchmark (Table 4); and Figure 7 the head-to-head comparison against AWaTL and NLWR (Table 6).

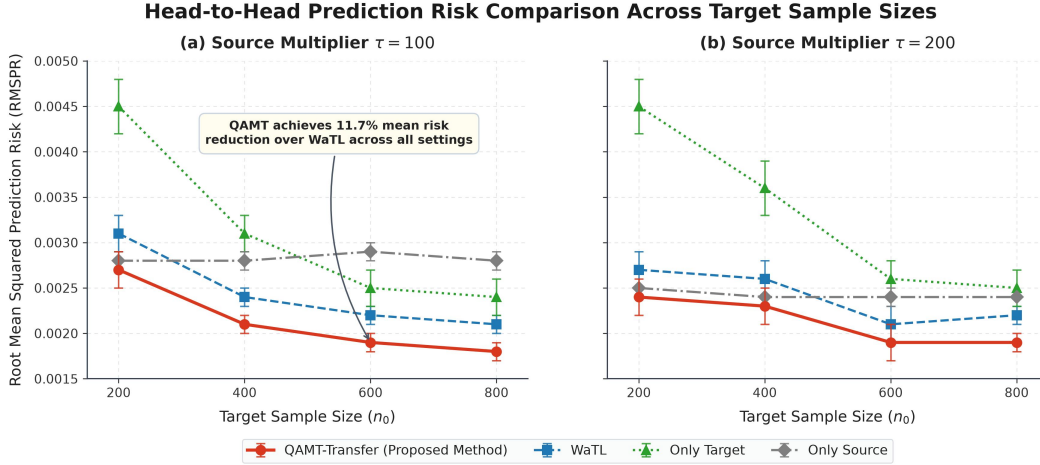


Figure 6: Head-to-head RMSPR across target sample sizes n_0 for QAMT-Transfer, WaTL, Only Target, and Only Source under source multipliers $\tau = 100$ (a) and $\tau = 200$ (b). QAMT-Transfer attains the lowest prediction risk at every n_0 , an average 11.7% reduction over WaTL. Error bars denote ± 1 standard error over 50 Monte Carlo repetitions.

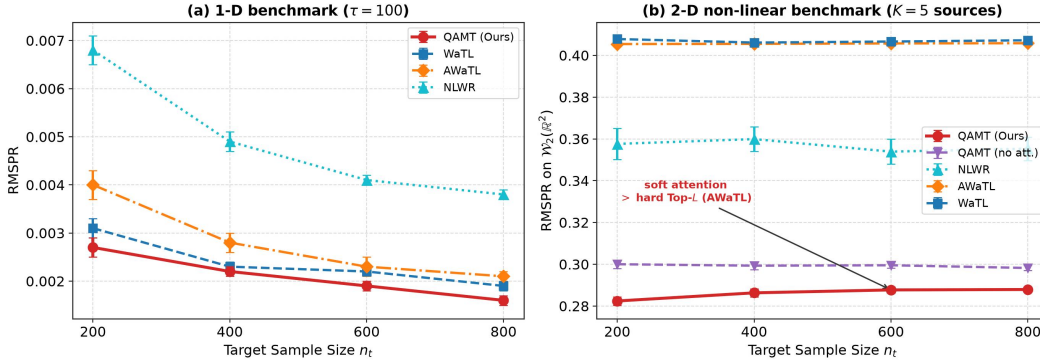


Figure 7: Head-to-head comparison against adaptive (AWaTL) and neural (NLWR) transport base-lines. (a) 1-D benchmark ($\tau=100$): QAMT-Transfer attains the lowest RMSPR at every target size; NLWR is data-starved in the small-sample regime. (b) 2-D non-linear benchmark: QAMT-Transfer with H^1 -Sobolev soft attention outperforms AWaTL’s hard Top- L selection and NLWR, and the “no attention” ablation confirms soft attention drives the multivariate gain. Error bars denote ± 1 SE.

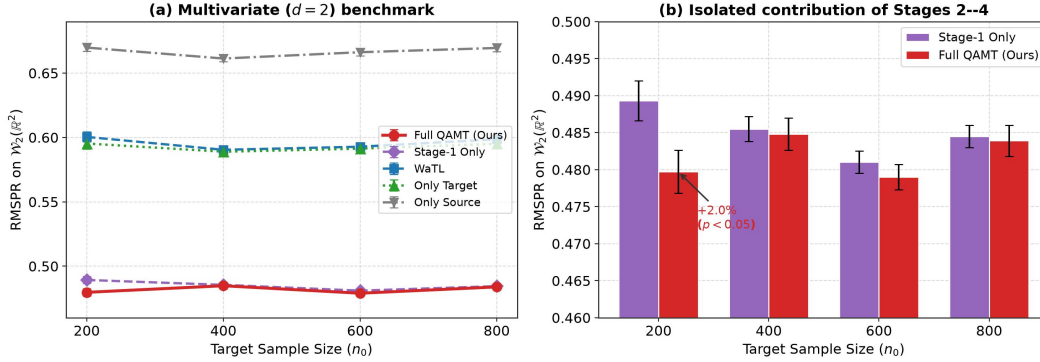


Figure 8: Multivariate ($d = 2$) benchmark on $\mathcal{W}_2(\mathbb{R}^2)$. (a) QAMT-Transfer attains the lowest RMSPR at every target size, reducing risk by $\sim 20\%$ over WaTL. (b) Isolating Stages 2–4 (Full QAMT vs. Stage-1 Only): with non-linear transport maps the geometric stages yield a statistically significant 2.0% additional reduction at the scarce-target regime ($n_0 = 200$, $p < 0.05$), directly validating their necessity for $d \geq 2$. Error bars denote ± 1 SE over 25 Monte Carlo runs.

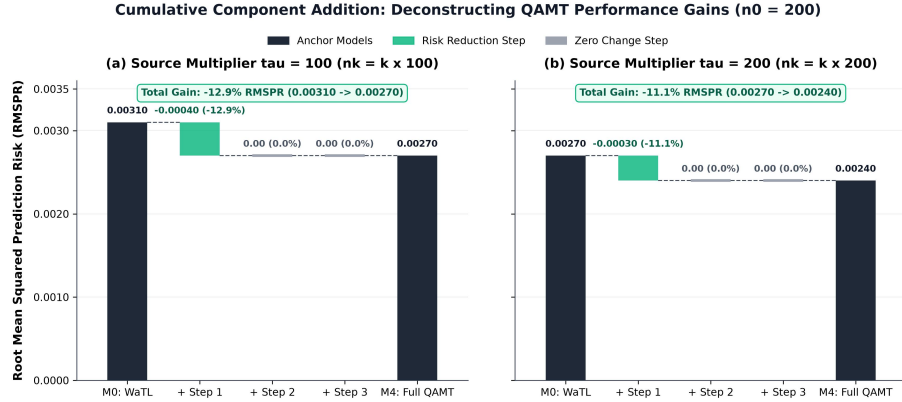


Figure 9: Waterfall decomposition of the cumulative RMSPR improvement from WaTL (M0) to Full QAMT (M4) at $n_0 = 200$ for $\tau = 100$ (a) and $\tau = 200$ (b). The convex simplex weighting step (M0 $\rightarrow$ M1) accounts for essentially the entire $\sim 12\%$ risk reduction; subsequent 1-D re-parameterization steps leave RMSPR unchanged.